

Testing Roughness in Realized Volatility

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- Volatility is one of the most important quantities in finance. It affects option prices, hedging, portfolio risk etc.
- Empirical volatility paths often look very irregular.
- This irregularity is usually summarized by a Hurst parameter H .
- Brownian motion roughness corresponds to $H \approx 0.5$.
- Rough-volatility models often use smaller values such as $H \approx 0.1 - 0.3$.

Main question

When volatility looks rough, we should ask whether this is a real market feature or only an effect of how volatility is measured.

Question: Is the roughness observed in realized volatility a genuine market feature, or is it introduced by the estimation procedure?

Spot and realized volatility

Spot volatility

A standard continuous-time log-price model is

$$dX_t = \mu_t dt + \sigma_t dB_t, \quad X_t = \log S_t.$$

The process σ_t is the **spot volatility**.

Meaning: it is the instantaneous volatility driving the stock-price path at time t .

Problem: σ_t is latent. We do not observe it directly from market prices.

Realized volatility

From observed log-returns

$$\Delta X_i = X_{t_{i+1}} - X_{t_i},$$

realized volatility over a window is

$$RV_{t,t+\Delta} = \left(\sum_{t_i \in [t, t+\Delta]} (\Delta X_i)^2 \right)^{1/2}.$$

Meaning: RV is an observable proxy constructed from the stock-price path.

Key point: RV is not the same object as σ_t .

Goal

Main goal

Test whether realized-volatility roughness reflects true spot-volatility roughness, or is affected by how realized volatility is computed.

- Simulate volatility paths with known H .
- Estimate $\hat{H}(\sigma)$ from true volatility.
- Compute realized volatility from price paths.
- Compare $\hat{H}(\sigma)$ with $\hat{H}(RV)$.
- Check sensitivity to RV window size and market data.

Key question

Can realized volatility alone justify rough spot volatility?

Roughness estimator

Estimator idea

The roughness is estimated by choosing the power p for which a normalized p -variation statistic behaves like the time clock. Then the estimated Hurst parameter is $\hat{H} = 1/\hat{p}$.

For observations $X_0, \dots, X_L$ on $[0, T]$, split the path into K coarse blocks. Define

$$W(p) = \sum_{i=0}^{K-1} \frac{|X_{b_{i+1}} - X_{b_i}|^p}{\sum_{j=b_i}^{b_{i+1}-1} |X_{j+1} - X_j|^p} (t_{b_{i+1}} - t_{b_i}).$$

Choose

$$\hat{p} \approx \arg \min_{p \in [1, 12]} |W(p) - T|, \quad \hat{H} = \frac{1}{\hat{p}}.$$

Interpretation

- $H \approx 0.5$: Brownian-like roughness.
- $H < 0.5$: rougher than Brownian motion.
- Smaller H means a more irregular path.

Experiment 1: fractional Brownian motion sanity check

Purpose

Before using the estimator on volatility, we first test it on paths whose roughness is known.

- Generate fractional Brownian motion paths B_t^H .
- Use Hurst parameters

$$H = 0.1, 0.3, 0.5, 0.8.$$

- Choose candidate powers

$$p \in [1, 12].$$

- Estimate $\hat{H}_{L,K}$ from each path.

Results

H	Min.	Mean	Std.	Max.
0.1	0.0833	0.1060	0.0201	0.1624
0.3	0.2460	0.2991	0.0197	0.3460
0.5	0.4693	0.5036	0.0117	0.5273
0.8	0.7379	0.7848	0.0206	0.8447

Table: Estimated roughness index $\hat{H}_{L,K}$ for fBm. The simulations use $N = 10000$, $K = 99$, and 50 independent paths.

Experiment 1: What does estimator do?

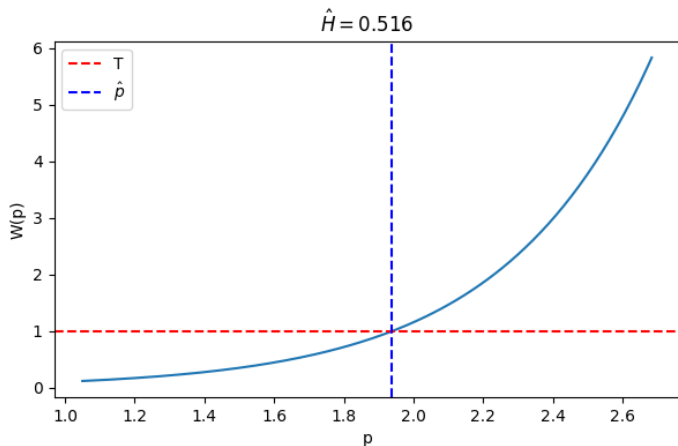


Figure: The estimated p is the intersection of $W(p)$ with the horizontal line $T = 1$.

Experiment 2: fractional OU volatility model

Model idea

We simulate a fractional Ornstein–Uhlenbeck stochastic-volatility model and compare the roughness estimated from the true instantaneous volatility with the roughness estimated from realized volatility.

Instantaneous volatility

For each value of H , we simulate

$$dY_t = -\gamma Y_t dt + \theta dB_t^H, \quad \sigma_t = \sigma_0 e^{Y_t}.$$

Here B_t^H is fractional Brownian motion, so the input roughness is known.

Stock price and realized volatility

Using σ_t , we simulate

$$dX_t = -\frac{1}{2}\sigma_t^2 dt + \sigma_t dW_t, \quad S_t = e^{X_t}.$$

Then we compute

$$RV_i = \left(\sum_{j=i-m+1}^i (\Delta X_j)^2 \right)^{1/2}, m = 10.$$

Estimator settings

We use $N = 90000$, $K = 300$, overlapping realized-volatility windows, and estimate $\hat{H}(\sigma)$ and $\hat{H}(RV)$.

Experiment 2: instantaneous vs realized volatility roughness

Estimated roughness indices

H	$\hat{H}(\sigma)$	$\hat{H}(RV)$
0.10	0.088	0.224
0.20	0.211	0.218
0.30	0.293	0.201
0.40	0.392	0.145
0.50	0.499	0.166
0.60	0.597	0.164
0.70	0.683	0.163
0.80	0.773	0.158

Main observation

The estimator recovers the roughness of the instantaneous volatility reasonably well:

$$\hat{H}(\sigma) \approx H.$$

However, the realized-volatility roughness remains small:

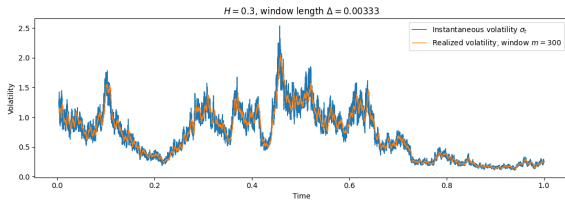
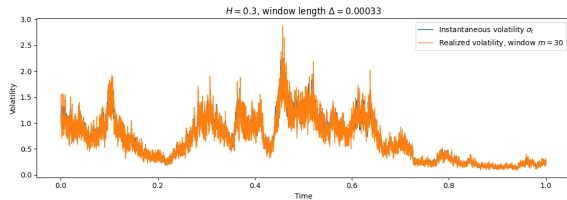
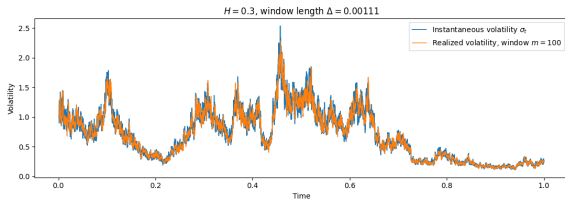
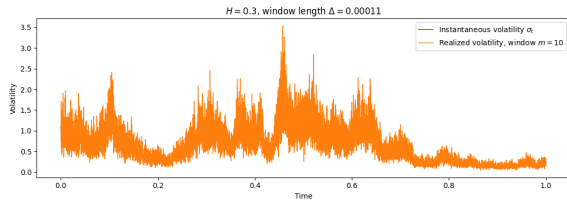
$$\hat{H}(RV) < 0.3$$

for all tested values of H .

Conclusion

Realized volatility can appear rough even when the underlying instantaneous volatility is not very rough.

Experiment 3: Effect of changing window size



Experiment 3: Effect of changing window size

Setup

We fix the true volatility roughness at $H = 0.3$ and use $N = 90000$, $K = 300$. For each window size m , realized volatility is computed from m consecutive log returns.

m	Δ	$\hat{H}(\sigma)$	$\hat{H}(RV)$
10	0.00011	0.29056	0.18523
30	0.00033	0.28049	0.32981
100	0.00111	0.28381	0.49889
300	0.00333	0.30167	0.65755

Observation

The estimate from instantaneous volatility remains close to $H = 0.3$, while the estimate from realized volatility changes strongly with the window size.

Market-data experiment: AAPL realized volatility

Data and construction

We download 1-minute AAPL close prices using `yfinance`, keep only regular trading hours, and compute log returns

$$r_i = \log S_{t_i} - \log S_{t_{i-1}}.$$

For a window of m minutes, realized volatility is computed as $RV_i = \left(\sum_{j=i-m+1}^i r_j^2 \right)^{1/2}$.

Downloaded data

- Start: 2026-06-10, 09:30
- End: 2026-07-08, 15:59
- Trading days: 19
- Price observations: 7410

Estimator step

For each realized-volatility window m , we obtain a path

$$RV_0, RV_1, \dots, RV_L.$$

Then we estimate roughness using the normalized p -variation estimator.

Market-data experiment: roughness across RV windows

Estimated roughness of AAPL realized volatility

RV window	Length of RV path	$\hat{H}(RV)$
2	7372	0.1132
5	7315	0.2168
10	7220	0.3267
15	7125	0.4258
30	6840	0.5025
60	6270	0.6313

Observation

The estimated roughness depends strongly on the realized-volatility window size:

$$m \uparrow \implies \hat{H}(RV) \uparrow.$$

Small windows give rougher realized-volatility paths, while larger windows smooth the path and increase the estimated Hurst index.

Takeaway

A single value of $\hat{H}(RV)$ can be misleading. The estimate should be reported together with the realized-volatility window size.

Conclusion

Main conclusion

Realized-volatility roughness is not a fixed property of the market path alone. It also depends strongly on how realized volatility is constructed.

Simulation evidence

- The estimator recovers the true roughness of instantaneous volatility: $\hat{H}(\sigma) \approx H$.
- But realized volatility can give much smaller or larger estimates depending on the RV window.

Market-data evidence

- For AAPL 1-minute data, $\hat{H}(RV)$ increases with the RV window.
- Small windows give rougher estimates.
- Larger windows smooth the RV path and increase $\hat{H}(RV)$.

Final note

A low estimate of $\hat{H}(RV)$ should not automatically be interpreted as proof that spot volatility is rough. The realized-volatility window size must be reported and tested carefully.

Thank You